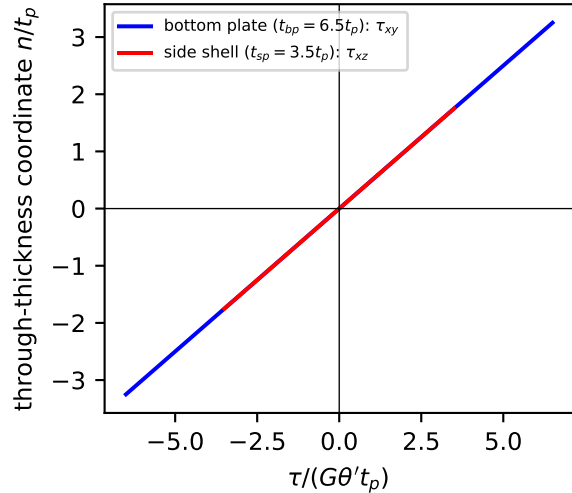
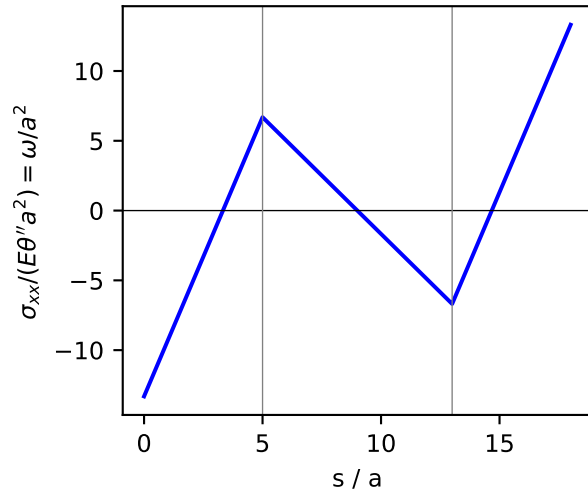


free warping (St. Venant) shear stress



constrained warping normal stress  $\sigma_{xx}(s)$  (and  $u(s) \propto \omega$ )



constrained warping shear stress  $\tau(s) = -E\theta'''S_\omega/t$

